

HW 1: Confidence Intervals for Means

1) A nutritionist wants to estimate the average amount of sugar in a certain brand of yogurt. She takes an SRS of 20 containers and measures the sugar content, in grams. She creates a 90% confidence interval for the true amount of sugar in this brand of yogurt: (11.42, 12.58).

- (a) What is the point estimate from this sample?
- (b) What is the margin of error?
- (c) Interpret the 90% confidence interval in the context of the problem.

- (d) Interpret the confidence level of 90% in the context of the problem.

2) Your local coffee shop brews their own coffee blend in their warehouse on site. They recently entered into an agreement with a national chain store to sell their coffee blend. Their contract with the store states that there will be an average of 30 bags of their coffee on the shelves at all times. The owner of the coffee shop decides to check if this is true by going around to the 6 local chain stores and records the number of their coffee bags on the shelves. She plans on constructing a 95% confidence interval to determine the mean number of coffee bags on display in all the chain stores.

- (a) Does the situation above meet each of the conditions for constructing a confidence interval? If a condition has not been met, explain how it would impact the confidence interval and what should be changed about this study to satisfy the condition.

(b) If, instead of constructing a 95% confidence interval, the business owner constructed a 98% confidence interval, would the 98% interval be wider, narrower, or the same width as the 95% interval? Explain.

(c) How would the width of confidence interval change if she took a larger sample? Explain.

3) A biologist is studying the growth of a certain type of plant. She measures the heights (in cm) of 12 randomly selected plants after 8 weeks of growth:

24, 26, 23, 25, 27, 26, 24, 25, 30, 21, 25, 30

Construct and interpret the 90% confidence interval.